Scaling Ensembles: Function-Space Similarity Across Parameter Count

Experimental notes generated from the `scaling-ensembles` repository

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Abstract

This report studies whether independently trained neural networks become more functionally similar as their parameter count increases. The starting point is Fort, Hu, and Lakshminarayanan's "Deep Ensembles: A Loss Landscape Perspective", which argues that deep ensembles work well because independent random initializations explore distinct modes in function space, while trajectory-local subspace methods tend to remain inside one prediction-space mode. Here we reproduce a small subset of that experimental logic, but add parameter count as the main scaling axis. We run width sweeps for an MNIST MLP, a CIFAR-10 CNN, and a CIFAR-10 patch-transformer classifier, then compare accuracy, pairwise prediction agreement, Jensen-Shannon divergence, ensemble gain, and linear interpolation barriers.

1 Background

Let a neural network with parameters θ define a predictive distribution $p_\theta(y \mid x)$ and a class prediction

$$f_\theta(x) = \arg\max_y p_\theta(y \mid x).$$

Deep ensembles train several networks from different random initializations, producing parameters $\theta_1, \dots, \theta_M$. The ensemble prediction is usually the arithmetic mean of predictive distributions:

$$p_{\text{ens}}(y \mid x) = \frac{1}{M} \sum_{m=1}^{M} p_{\theta_m}(y \mid x).$$

If the ensemble members make accurate but decorrelated errors, averaging can improve both accuracy and uncertainty estimates.

The key distinction in Fort et al. is between *weight-space* similarity and *function-space* similarity. Two models may be far apart in weight space but compute similar functions, or they may be connected by a low-loss path while still representing substantially different classifiers. The paper therefore measures not only loss or parameter distance, but also prediction-space quantities such as disagreement, Jensen-Shannon divergence, and t-SNE embeddings of softmax vectors.

2 Connection to Fort et al.

Fort et al. perform several experiments that motivate this project:

- compare checkpoints within one training trajectory against checkpoints from different random initializations;

- visualize prediction vectors with t-SNE;
- sample local subspaces around one optimum using dropout, diagonal Gaussian approximations, low-rank Gaussian approximations, and random subspaces;
- build diversity-accuracy plots comparing independent initializations to subspace samples;
- compare baseline models, subspace sampling, deep ensembles, and ensemble-plus-subspace methods;
- test behavior under dataset shift on CIFAR-10-C and ImageNet-C.

This report focuses on a narrower question: when architecture width and parameter count increase, do independently trained minima become more similar as functions? This is related to the paper's diversity-accuracy plane, but uses parameter count as an explicit independent variable.

3 Metrics

For a pair of independently trained models a, b, the experiments compute:

$$\text{agreement}(a, b) = \frac{1}{N}\sum_{n=1}^{N} \mathbf{1}\{f_{\theta_a}(x_n) = f_{\theta_b}(x_n)\}, \tag{1}$$

$$\text{disagreement}(a, b) = 1 - \text{agreement}(a, b). \tag{2}$$

The Jensen-Shannon divergence is computed from the predictive distributions:

$$\text{JS}(p_a, p_b) = \frac{1}{2}\text{KL}(p_a \| m) + \frac{1}{2}\text{KL}(p_b \| m), \quad m = \frac{1}{2}(p_a + p_b).$$

We also compute a two-member ensemble accuracy for each pair and subtract the mean individual accuracy to estimate an *ensemble gain*. Finally, to probe loss-landscape connectivity, we linearly interpolate in parameter space:

$$\theta(\alpha) = (1 - \alpha)\theta_a + \alpha\theta_b, \quad \alpha \in [0, 1],$$

and report the maximum loss barrier relative to the worse endpoint loss.

4 Experimental Setup

All runs used PyTorch with Apple's Metal backend (`mps`) exposed to the process. The experiments are pilot-scale: they use dataset subsets and three random seeds per width. This is sufficient for code validation and trend finding, but not for statistical claims.

5 Results

5.1 MNIST MLP

The MNIST run gives the cleanest partial support for the scaling hypothesis. Accuracy rises monotonically, and interpolation barriers shrink substantially as parameter count increases. However, prediction-space similarity is not monotonic: agreement peaks and JS divergence is lowest at width 128.

Experiment	Dataset	Architecture	Widths
MNIST MLP	MNIST	2-hidden-layer MLP	32, 64, 128, 256
CIFAR-10 CNN	CIFAR-10	Small convolutional net	16, 32, 64, 128
CIFAR-10 Patch Transformer	CIFAR-10	ViT/DiT-style patch classifier	64, 128, 256

Table 1: Pilot experiments. Each width was trained with three random seeds.

Width	Params	Acc.	Agreement	JS	Barrier
32	26.5k	88.12%	93.00%	0.0174	1.504
64	55.1k	89.12%	94.32%	0.0144	0.979
128	118.3k	90.93%	95.17%	0.0118	0.884
256	269.3k	91.67%	93.67%	0.0207	0.784

Table 2: MNIST MLP results. Accuracy improves with parameter count. Function similarity improves until width 128, then weakens at width 256.

5.2 CIFAR-10 CNN

Width	Params	Acc.	Agreement	JS	Barrier
16	38.6k	50.62%	71.35%	0.0291	0.660
32	151.9k	51.83%	69.98%	0.0418	0.770
64	602.8k	60.17%	72.48%	0.0494	0.984
128	2.40M	62.47%	74.92%	0.0451	1.155

Table 3: CIFAR-10 CNN results. Accuracy and agreement rise at larger width, but linear interpolation barriers also rise.

The CNN sweep suggests that accuracy scaling and mode connectivity need not move together. Larger CNNs are more accurate and modestly more agreement-heavy, but the mean maximum interpolation barrier increases. In this pilot, wider CNNs find better optima without becoming more linearly connected.

5.3 CIFAR-10 Patch Transformer

The patch transformer contradicts the simplest version of the hypothesis in this pilot. Agreement decreases and JS divergence increases with parameter count. The likely explanation is undertraining: transformer-style classifiers usually need stronger optimization, augmentation, longer schedules, and often larger data regimes than a small CNN.

6 Paper-Style Plots

Figure 1 repeats the spirit of Fort et al.'s diversity-accuracy plane, but labels each point with width and parameter count. In that paper, independent initializations occupy the high-diversity/high-accuracy frontier relative to subspace samples. In our pilot, model family matters strongly: MNIST points live at high accuracy and low disagreement, CIFAR CNN points move toward better accuracy with moderate disagreement, and patch-transformer points show high disagreement but poor accuracy.

Width	Params	Acc.	Agreement	JS	Barrier
64	208.1k	44.08%	52.37%	0.0646	0.641
128	809.4k	44.43%	46.87%	0.0851	0.619
256	3.19M	40.57%	41.08%	0.0928	0.689

Table 4: CIFAR-10 patch-transformer results. In this short run, larger transformers are less similar in function space and less accurate.

7 Interpretation

The current evidence supports a more nuanced hypothesis than "more parameters always imply more similar functions." Parameter count interacts with dataset, architecture, and optimization quality.

- On MNIST MLP, larger models become more accurate and more linearly connected, but function-space similarity is non-monotonic.
- On CIFAR-10 CNN, larger models become more accurate and somewhat more similar in predictions, but linear barriers increase.
- On CIFAR-10 patch transformer, larger models are more diverse but not more accurate, indicating that this architecture is undertrained in the current pilot.

8 Next Experiments

The next version should move closer to the Fort et al. setup:

1. Use at least 10–20 random seeds per width to estimate confidence intervals.
2. Train to matched endpoint train loss, not just fixed epochs.
3. Add CIFAR-10-C and SVHN to evaluate dataset shift and OOD diversity.
4. Add trajectory checkpoints and t-SNE plots of flattened softmax predictions.
5. Add subspace sampling around each optimum: dropout, diagonal Gaussian, low-rank Gaussian, and random directions.
6. For the transformer, add longer schedules, augmentation, and possibly AdamW warmup/cosine decay before drawing conclusions.

9 Conclusion

The experiments reproduce several measurement ideas from "Deep Ensembles: A Loss Landscape Perspective" while making parameter count explicit. The most important observation is that function-space similarity does not scale universally with parameter count. Scaling can reduce loss barriers, improve accuracy, or increase diversity depending on architecture and optimization. This suggests a promising paper direction: characterize when overparameterized minima converge to similar functions and when they remain diverse enough for deep ensembles to keep providing gains.

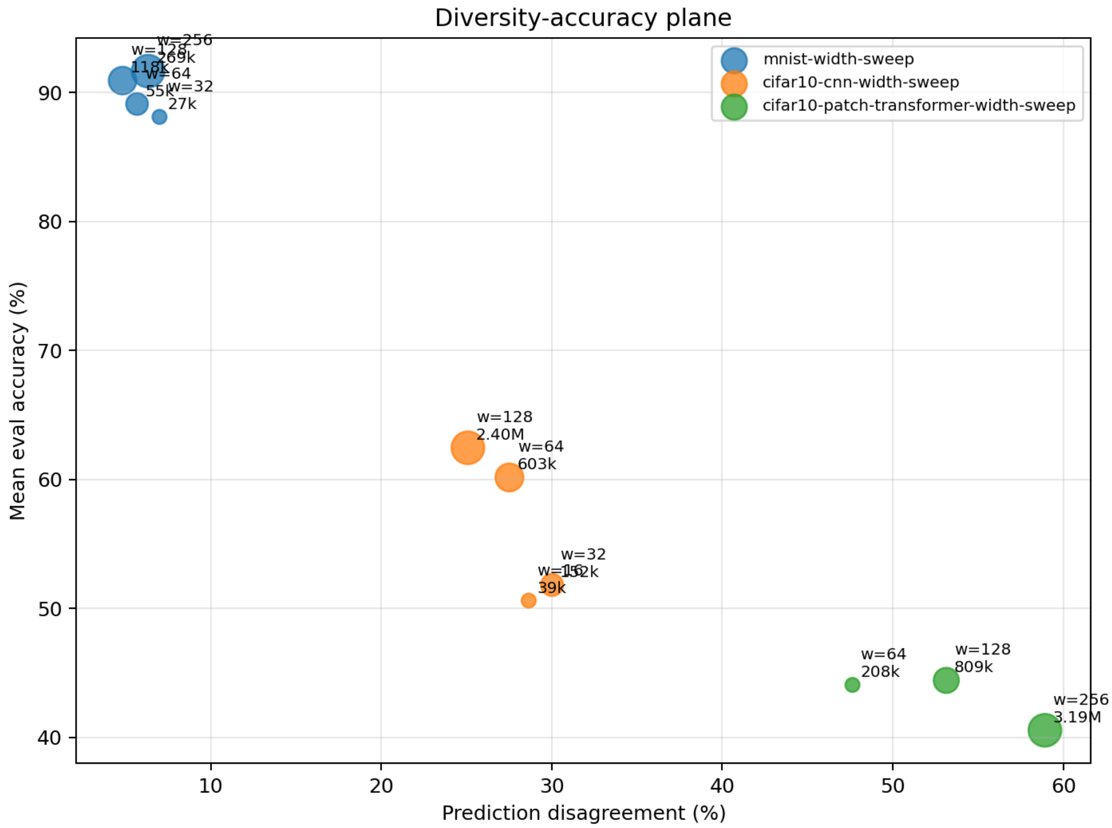

Figure 1: Diversity-accuracy plane with marker size and labels tied to parameter count.

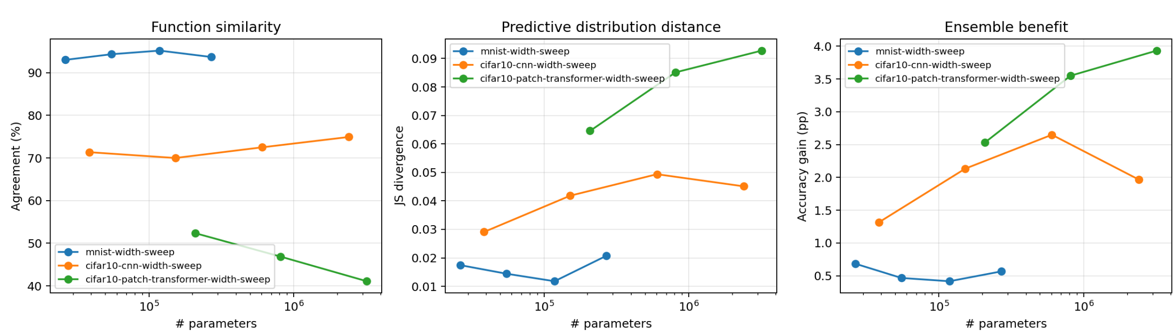

Figure 2: Function similarity, predictive divergence, and ensemble gain as a function of parameter count.

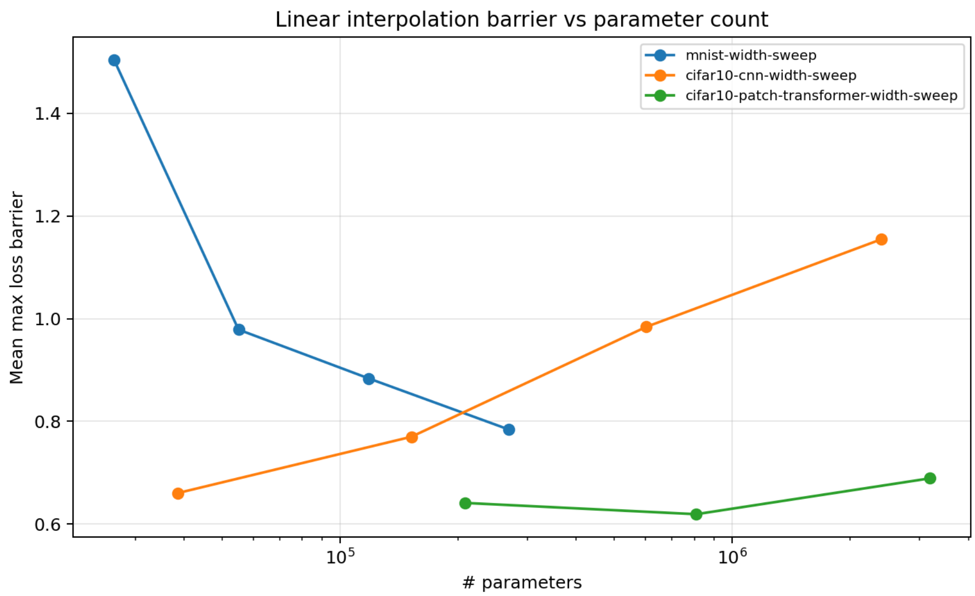

Figure 3: Mean maximum linear interpolation barrier versus parameter count.